

Project Design Phase-II
Data Flow Diagram & User Stories

Date	30 July 2026
Team ID	Panchumarthi Durga Kosali
Project Name	Metro Ticket Generating System in ServiceNow
Maximum Marks	4 Marks

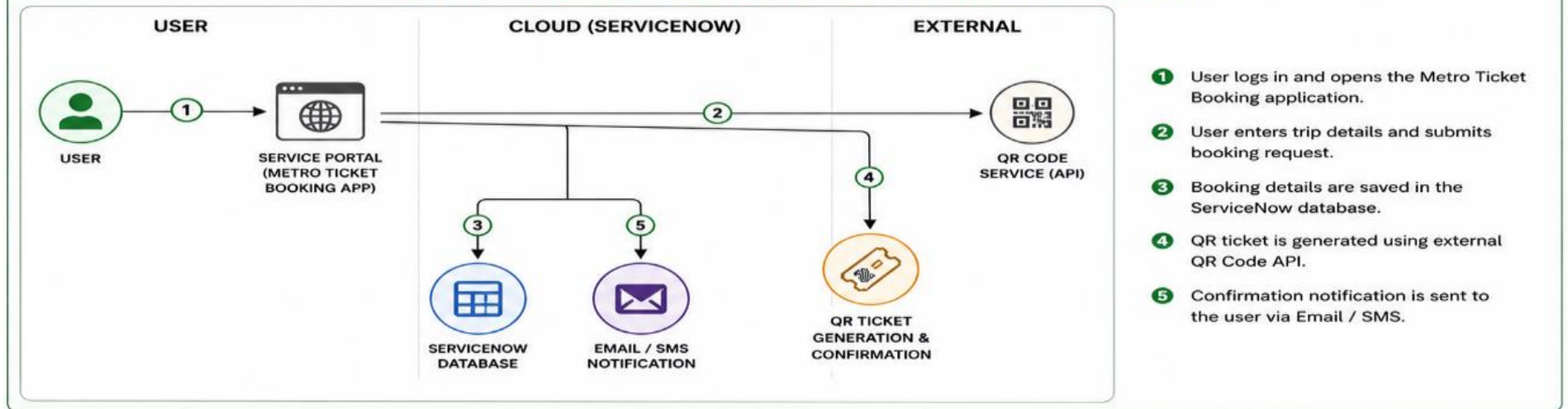
Data Flow Diagrams:

A Data Flow Diagram (DFD) is a graphical representation that illustrates how data moves through the **Metro Ticket Generating System in ServiceNow**. It provides a clear understanding of the flow of information between the passenger, Service Portal, ServiceNow platform, database, and QR ticket generation module. The DFD represents how user inputs are collected, processed, validated, stored, and transformed into meaningful outputs.

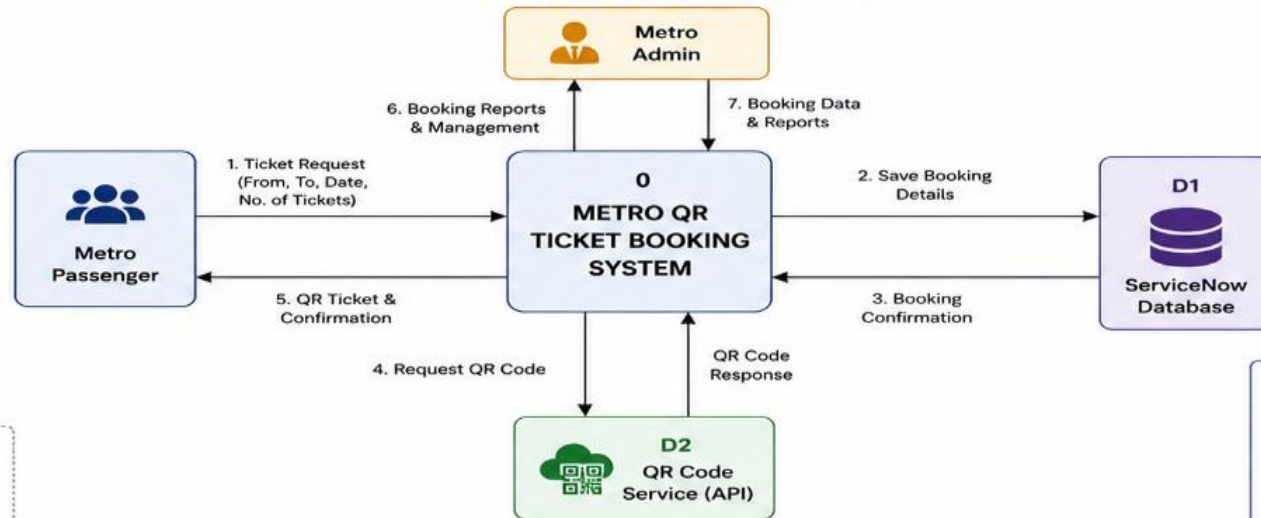
In this system, the passenger accesses the Service Portal and enters journey details such as the source station, destination station, ticket type (Single/Return), number of passengers, and payment details. These inputs are sent to the ServiceNow application, where the fare is calculated automatically based on the selected route and ticket type. After successful validation and payment processing, the booking information is securely stored in the Metro Ticket Database.

The system then generates a unique QR code containing the ticket details, which is displayed to the passenger and can be scanned at the metro entry gate for authentication. The DFD also demonstrates how data is exchanged between different processes, ensuring secure communication, accurate fare calculation, reliable ticket generation, and efficient record management. Overall, the Data Flow Diagram provides a structured view of the complete ticket booking process, making it easier to understand the system architecture, data movement, and interactions among various components.

FLOW – METRO QR TICKET BOOKING SYSTEM



DATA FLOW DIAGRAM (DFD) – LEVEL 0 METRO QR TICKET BOOKING SYSTEM



LEGEND



DATA STORES

- D1 : ServiceNow Database**
(Stores booking, user and transaction data)
- D2 : QR Code Service (API)**
(External service to generate QR ticket)

User Stories

User Type	Functional Requirement (Epic)	User Story Number	User Story / Task	Acceptance criteria	Priority	Release
Metro Passenger	Login	USN-1	As a user, I can register for the application by entering my email, password, and confirming my password.	User can access the Metro Ticket Booking page.	High	Sprint-1
Metro Passenger	Ticket Booking	USN-2	As a user, I will receive confirmation email once I have registered for the application	Ticket request is created successfully.	High	Sprint-1
Metro Passenger	Fare Calculation	USN-3	As a user, I can register for the application through Facebook	Correct fare is displayed automatically.	High	Sprint-1
Metro Passenger	QR Ticket Generation	USN-4	As a user, I can register for the application through Gmail	QR code is displayed successfully.	High	Sprint-1
Metro Passenger	Download Ticket	USN-5	As a user, I can log into the application by entering email & password	QR ticket is downloaded successfully.	Medium	Sprint-2
Metro Admin	Booking Management	USN-6	As an admin, I can view all metro ticket bookings.	All booking records are visible in the dashboard.	High	Sprint-1
Metro Admin	Request Approval	USN-7	As an admin, I can manage ticket requests.	Requests can be viewed and updated.	Medium	Sprint-2